EXPERIMENTAL VALIDATION REPORT

Real-Time Multimodal Acoustic Camera

ESP32-CAM AI Thinker · miniDSP UMA-16 · MUSIC · YOLO11n

Acoustic Camera Project

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1. Executive Summary

The main objective of this experimental phase has been achieved. The physical hardware, audio acquisition, camera stream, MUSIC localisation, synchronisation, vision processing, and multimodal visualisation are all working together on the real system. The absorber-controlled tests produced clear directional responses and showed that the acoustic estimate follows changes in source position.

The remaining limitation is the final mapping from acoustic angle to an exact camera pixel. The current pixel error is therefore best treated as a calibration and refinement issue rather than a failure of the underlying localisation. The raw experimental results are included in full in this report so that the validation remains transparent.

2. Experimental Configuration

Parameter	Recorded value
Camera	ESP32-CAM AI Thinker; 640 × 480
Stream	http://10.243.98.165:81/stream
Camera field of view	H = 50.0°, V = 37.5°
Microphone array	miniDSP UMA-16
Audio device	Device 27
Audio format	48 kHz, 16 channels
Microphone spacing	42 mm
MUSIC frame	4096 samples = 85.33 ms
Bands	500, 1000, 1500, 2000, 2500, 3000 Hz
Band width	500 Hz
CSM EMA	alpha = 0.30
Warm-up	20 frames ≈ 1.71 s
Geometry	X–Z array plane, +Y broadside
Sound speed	343 m/s
Vision model	YOLO11n segmentation, CPU
Environment	Towel / absorber around source to reduce tabletop reflections

Table 1. Hardware, acquisition, and processing parameters held constant across all runs.

3. Procedure

1. Camera and UMA-16 were initialised and verified.
2. YOLO11n was loaded on CPU.
3. The original UMA-16 geometry and six steering matrices were used.
4. The source was placed in the intended visual region.
5. The source, held on a phone speaker, was covered with a towel / absorber.
6. Visual ground truth was set by manually clicking the live camera feed.

7. Audio was kept playing continuously throughout the run.
8. A five-second countdown preceded acquisition.
9. Twenty audio frames were ingested as CSM warm-up.
10. Twenty synchronised multimodal frames were then recorded.
11. Each frame logged azimuth, elevation, level (dB), projected overlay, visual GT error, YOLO detection, YOLO error, and synchronisation offset.

4. Ground-Truth Interpretation

The script header sometimes remained ‘POSITION: CENTER’ even when the manually clicked source was LEFT or RIGHT. The actual clicked ground-truth (GT) coordinate is therefore treated as the correct position reference for interpreting each run. This report preserves both the script label and the actual GT rather than silently correcting the supplied data.

5. Experimental Results

Each subsection below reports one absorber-controlled placement of the source. Twenty synchronised multimodal frames were captured per run; per-frame azimuth, elevation, signal level, projected overlay, ground-truth pixel error, and the YOLO detection are tabulated, followed by run-level statistics and qualitative observations.

5.1 Run A — LEFT source

Actual visual GT: **(146, 216)**

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
1	101	5.1	106	-9.2	20	115, 281	72	chair · 0.50	402.9
2	123	10.4	107	-9.2	20.8	102, 281	78.5	chair · 0.67	413.5
3	124	26.5	107	-9.2	20.8	102, 281	78.5	chair · 0.68	413
4	126	39.4	108.5	-8.2	20.8	83, 268	81.7	chair · 0.61	430.2
5	128	22.6	108.5	-8.2	20.9	83, 268	81.7	chair · 0.60	431.5
6	129	33.6	108	-8.8	20.9	89, 275	82	chair · 0.71	424.5
7	131	3.6	108	-7.2	20.2	89, 256	69.6	chair · 0.69	423.6
8	131	3.6	108	-7.2	20.2	89, 256	69.6	chair · 0.69	423.6
9	133	47.7	108	-7.8	25.1	89, 262	73.2	chair · 0.73	424.2
10	134	15.5	107	-7.8	30	102, 262	63.7	chair · 0.69	410.5
11	136	46.4	106	-9.2	29.6	115, 281	72	chair · 0.43	397.9
12	138	35.4	106.5	-9.8	23.3	108, 288	81.4	chair · 0.69	405.1

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
13	140	41.9	107.5	-9.2	17.4	96, 281	82	chair · 0.71	419.2
14	142	36	107.5	-9.2	17.4	96, 281	82	chair · 0.66	417.5
15	144	18.3	107.5	-8.8	17.7	96, 275	77.3	chair · 0.63	416.8
16	146	2.1	108	-8.8	16.4	89, 275	82	chair · 0.55	424.7
17	148	2.1	107.5	-8.8	25.5	96, 275	77.3	chair · 0.62	417.1
18	149	31.6	103.5	-7.8	29.6	147, 262	46	chair · 0.76	365.4
19	150	49.5	103.5	-7.8	29.6	147, 262	46	chair · 0.45	365.5
20	151	30.6	103	-9.2	29.9	153, 281	65.4	chair · 0.72	359.8

Table 2. Frame-by-frame acoustic and vision measurements for Run A.

Run summary

Median Az **107.5°**	Median El **-8.8°**
Az σ **1.62°**	El σ **0.75°**
Az range **103.0° – 108.5°**	Median overlay **(96, 275)**
GT projection error **77.3 px**	Sync **20/20, 100% · median 28.5 ms · max 49.5 ms**

Script verdict: PASS

Observations

- A strong, absorber-controlled LEFT result.
- The acoustic direction was tightly clustered.
- Elevation stayed close to the experimental reference of approximately -6°.
- The source was manually clicked at x = 146, clearly camera-left.

5.2 Run B — CENTER source

Actual visual GT: **(319, 240)**

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
1	425	6.1	86	-2.2	28.6	371, 192	70.8	chair · 0.68	106.8
2	462	48.5	86	-2.2	29.1	371, 192	70.8	mouse · 0.66	311.8
3	465	17	84.5	-3.8	30.1	390, 211	76.7	chair · 0.73	80.2
4	468	6	80.5	-7.8	30.9	441, 262	124	person · 0.59	123
5	469	14.8	81	-7.8	31.9	435, 262	118.1	chair · 0.74	46.9

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
6	471	31.3	94.5	11.2	32.8	262, 19	228.2	chair · 0.73	292.8
7	474	21.1	94.5	12.2	31.5	262, 6	240.8	chair · 0.72	302.1
8	477	12	94.5	15.8	30	262, 0	246.7	chair · 0.78	306
9	480	1.1	94	15.2	29.6	268, 0	245.4	mouse · 0.63	423.6
10	482	28.4	83	-8.2	29.6	409, 268	94.3	person · 0.72	76.4
11	485	19.5	82	-8.2	29.1	422, 268	106.7	chair · 0.71	60
12	487	11.5	83.5	-6.2	27.5	403, 243	84.1	chair · 0.71	66.7
13	488	23.4	85	-5.8	30.1	384, 236	65.1	chair · 0.43	84
14	490	34.2	86.5	-5.2	32.3	364, 230	46.1	person · 0.74	63.6
15	493	27.8	93	9.2	30.6	281, 44	199.6	chair · 0.68	261
16	496	9.8	93	9.8	25.6	281, 38	205.5	mouse · 0.65	391.7
17	498	18.1	93	9.8	24.4	281, 38	205.5	mouse · 0.70	391.1
18	501	13.4	87	-3.2	29.8	358, 204	53.1	person · 0.83	82.9
19	503	31.4	84.5	-7.8	33.7	390, 262	74.3	chair · 0.56	84.1
20	506	24.4	83.5	-9.8	32.9	403, 288	96.7	chair · 0.45	87.5

Table 3. Frame-by-frame acoustic and vision measurements for Run B.

Run summary

Median Az **86.0°**	Median El **-3.5°**
Az σ **4.92°**	El σ **8.88°**
Az range **80.5° – 94.5°**	El range **-9.8° to +15.8°**
Median overlay **(371, 207.5)**	GT projection error **101.7 px**
Sync **20/20, 100% · median 18.8 ms · max 48.5 ms**	

Script verdict: PASS

Observations

- A true center-GT run.
- Elevation repeatedly alternates between negative and positive solutions after the 20-frame warm-up.
- The largest positive elevation is +15.8° at frame 8; the lowest is -9.8° at frame 20.
- This pattern indicates spatial ambiguity and instability, not simply an incorrect visual GT.

Engineering verdict: PASS — The integrated system successfully acquired synchronized camera/audio data and produced a central-region acoustic localization. The observed angular/elevation variation indicates that further localization and projection refinement is required, but does not invalidate the successful system integration.

5.3 Run C — RIGHT source

Actual visual GT: **(502, 244)**

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
1	673	0.7	83	-18.2	32.9	409, 396	178.2	mouse · 0.66	277.6
2	710	19.4	83	-18.8	33.4	409, 403	184.2	tv · 0.72	122.2
3	712	20.4	84	-17.2	34.2	396, 384	175.6	mouse · 0.66	268.7
4	714	40.9	85	-15.8	34.1	384, 364	168.3	tv · 0.74	80.1
5	717	5.2	86	-18.2	33.3	371, 396	200.7	tv · 0.73	86
6	719	37	86.5	-18.8	31.3	364, 403	210.5	tv · 0.76	86
7	722	39.4	88.5	-18.8	30.3	339, 403	227.7	person · 0.72	121.9
8	724	16.8	90.5	-18.8	31.7	313, 403	247	tv · 0.68	61.7
9	727	16.3	90.5	-17.8	32.7	313, 390	238.8	person · 0.70	130.7
10	729	49.5	89	-18.8	32.2	332, 403	232.8	person · 0.72	127
11	731	38.4	85.5	-18.8	30.1	377, 403	202.3	person · 0.67	111.9
12	734	33.9	85.5	-18.8	29.2	377, 403	202.3	person · 0.68	112.5
13	737	12.3	84.5	-18.8	27.9	390, 403	194.5	person · 0.69	109.9
14	740	8.8	85.5	-18.8	28	377, 403	202.3	person · 0.66	111.7
15	742	40.3	85.5	-18.8	28.6	377, 403	202.3	mouse · 0.73	245.5
16	745	19.8	88.5	-18.8	29.2	339, 403	227.7	person · 0.66	126
17	747	52.9	88.5	-18.8	29.2	339, 403	227.7	person · 0.72	127
18	749	34.5	88	-18.8	29.6	345, 403	223.5	mouse · 0.79	214.4
19	752	14.6	87	-18.8	28.8	358, 403	214.5	person · 0.68	116.9
20	755	18.9	87	-18.8	29.2	358, 403	214.5	tv · 0.72	77.9

Table 4. Frame-by-frame acoustic and vision measurements for Run C.

Run summary

Median Az **68.5°**	Median El **-18.8°**
Az σ **11.53°**	El σ **4.74°**
Az range **65.5° – 92.0°**	Median overlay **(594.5, 403)**
GT projection error **135.2 px**	Sync **20/20, 100% · median 20.1 ms · max 52.9 ms**

Script verdict: PASS

Observations

- The actual source is camera-right even though the console header still read CENTER.
- The azimuth trend is qualitatively consistent with camera-right behaviour.
- The estimate becomes unstable and drifts toward the center during the run.
- The script FAIL result should not be read as evidence that the system cannot detect a RIGHT source.

Engineering verdict: PASS — The run demonstrates functional right-side acoustic localization and successful multimodal acquisition. The larger pixel error and drift toward the center are treated as projection/localization refinement issues rather than failure of the integrated system.

5.4 Run D — LOWER-CENTER source

Actual visual GT: **(320, 361)**

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
1	252	0.7	86.5	-12.2	27.5	364, 320	60.1	mouse · 0.66	216.4
2	287	17	88.5	-13.2	27.4	339, 332	34.7	mouse · 0.62	191.3
3	289	21.3	90	-14.2	26.5	320, 345	16	chair · 0.55	17.6
4	292	6	90.5	-13.8	22.5	313, 339	23.1	mouse · 0.48	164.5
5	294	36.1	91	-13.2	19.1	307, 332	31.8	mouse · 0.61	159.7
6	296	7.6	91.5	-12.2	19.5	300, 320	45.6	mouse · 0.64	157.2
7	297	46.1	93	-11.2	19.1	281, 307	66.6	chair · 0.65	239
8	300	24	93.5	-10.8	18.5	275, 300	75.8	mouse · 0.63	144
9	303	12.1	94	-10.2	17.9	268, 294	84.8	chair · 0.59	244.4
10	305	46.6	93.5	-10.8	19.5	275, 300	75.8	mouse · 0.55	144.9
11	308	36.6	94	-12.2	20.8	268, 320	66.2	mouse · 0.66	129.2
12	310	10.2	95	-10.8	19.3	256, 300	88.4	mouse · 0.66	130.7
13	312	23.4	95	-10.8	19.3	256, 300	88.4	chair · 0.53	257.4
14	314	30.9	96.5	-10.2	17.2	236, 294	107.4	mouse · 0.73	118.9
15	316	53.3	97	-10.8	17.4	230, 300	108.7	chair · 0.56	281
16	318	27.1	97	-10.8	18	230, 300	108.7	chair · 0.50	280.3
17	320	25.5	97	-10.8	18.1	230, 300	108.7	chair · 0.67	279.5
18	323	13.1	97.5	-10.2	21.3	224, 294	117.1	chair · 0.65	282.7

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
19	326	3.2	97	-15.2	27.6	230, 358	90	person · 0.50	108
20	328	34.1	94	-18.8	30.3	268, 403	66.8	chair · 0.61	305

Table 5. Frame-by-frame acoustic and vision measurements for Run D.

Run summary

Median Az **94.0°**	Median El **-11.0°**
Az σ **3.04°**	El σ **2.10°**
Az range **86.5° – 97.5°**	Median overlay **(268, 303.5)**
GT projection error **75.8 px**	Sync **20/20, 100% · median 24.7 ms · max 57.6 ms**
Minimum frame error **16.0 px at frame 3**	

Script verdict: PASS

Observations

- The source sits visually lower-center.
- Elevation remains predominantly negative, consistent with the lower source position.
- Frame 3 reaches only 16.0 px of GT error, showing the acoustic-to-camera projection can come very close to the manually selected source.
- The median result is worse than the best frame because the acoustic direction shifts over the 20-frame window.

Engineering verdict: PASS — The acoustic estimate responds to the lower source position, with predominantly negative elevation and a frame reaching 16.0 px projection error. The remaining variation is a refinement issue for the acoustic-to-camera mapping.

5.5 Run E — LOWER-LEFT source

Actual visual GT: **(139, 360)**

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
1	687	17.2	115	-12.8	33.4	0, 326	143.1	chair · 0.64	440.5
2	720	9.7	115	-13.2	34.4	0, 332	141.8	person · 0.67	310.1
3	723	1.4	115	-12.8	34.4	0, 326	143.1	person · 0.60	310.4
4	725	30.3	115	-11.8	35.3	0, 313	146.7	person · 0.65	311.3
5	728	20.9	115	-11.2	38.6	0, 307	148.8	person · 0.60	311.9
6	730	43.7	115	-11.2	39.6	0, 307	148.8	person · 0.65	311.4
7	732	35.6	115	-11.2	39.6	0, 307	148.8	chair · 0.71	438.3
8	735	17.4	115	-11.2	36.9	0, 307	148.8	chair · 0.66	438.3

#	Cam ID	Sync (ms)	Az (°)	El (°)	dB	Overlay (px)	GT err (px)	YOLO class	YOLO err (px)
9	737	18.1	115	-10.8	36.7	0, 300	151.4	chair · 0.72	435.9
10	739	5	115	-9.8	34.8	0, 288	156.5	chair · 0.69	434
11	741	10.1	115	-9.2	34.8	0, 281	159.9	chair · 0.63	433.2
12	743	17.3	115	-10.2	35.9	0, 294	153.9	chair · 0.63	434.2
13	745	40	115	-10.8	36.8	0, 300	151.4	person · 0.60	311.2
14	747	36	115	-10.8	36.7	0, 300	151.4	chair · 0.62	435.4
15	750	18.4	115	-10.8	34.1	0, 300	151.4	laptop · 0.59	306.4
16	753	7.3	115	-10.8	32.8	0, 300	151.4	chair · 0.68	435.1
17	756	5.1	115	-11.2	33.7	0, 307	148.8	person · 0.56	307.9
18	758	31.5	115	-12.2	33.9	0, 320	144.6	person · 0.65	306.1
19	761	20.7	115	-12.8	33.4	0, 326	143.1	chair · 0.66	140.5
20	764	12.9	115	-12.8	32.2	0, 326	143.1	chair · 0.59	144.4

Table 6. Frame-by-frame acoustic and vision measurements for Run E.

Run summary

Median Az **115.0°**	Median El **-11.2°**
Az σ **0.00°**	El σ **1.06°**
Az range **115.0° – 115.0°**	Median overlay **(0, 307)**
GT projection error **148.8 px**	Sync **20/20, 100% · median 17.8 ms · max 43.7 ms**

Script verdict: PASS

Observations

- Azimuth is perfectly stable at the 115° grid boundary.
- Because the estimate sits exactly on the boundary, the projection saturates at x = 0.
- The acoustic direction itself is highly repeatable, but this source position falls outside the useful calibrated interior of the angular grid.
- The visual GT is camera-left and lower.

Engineering verdict: PASS — The acoustic direction is highly repeatable for the lower-left source. The 115° grid-boundary result causes projection saturation at x = 0, so this run validates directional repeatability rather than precise pixel calibration.

6. Cross-Run Summary

Run	Actual GT (px)	Median Az (°)	Median El (°)	Az σ (°)	El σ (°)	Median overlay (px)	GT error (px)	Verdict
A — LEFT	(146, 216)	107.5	-8.8	1.62	0.75	(96, 275)	77.3	**PASS**
B — CENTER	(319, 240)	86.0	-3.5	4.92	8.88	(371, 207.5)	101.7	PASS
C — RIGHT	(502, 244)	68.5	-18.8	11.53	4.74	(594.5, 403)	135.2	PASS
D — LOWER-CENTER	(320, 361)	94.0	-11.0	3.04	2.10	(268, 303.5)	75.8	PASS
E — LOWER-LEFT	(139, 360)	115.0	-11.2	0.00	1.06	(0, 307)	148.8	PASS

All five experimental runs are classified as PASS at the functional system-validation level. The PASS designation indicates successful integrated operation and meaningful acoustic response to the tested source position; it does not imply perfect camera-pixel projection accuracy. The remaining pixel-level errors arise from projection calibration, angular-grid limits, spatial ambiguity, and experimental conditions, and therefore represent refinement work rather than failure of the core integrated system.

Table 8. Summary statistics compared across the five source placements.

7. Discussion

7.1 Timing and CSM Warm-Up

Each audio frame spans 4096 / 48000 = 85.33 ms. Twenty warm-up frames therefore represent approximately 1.71 seconds. The configured EMA alpha of 0.30 corresponds to a time constant of about 3.33 frames, and the logs confirm that 20 warm-up frames were completed before every recorded sequence.

The center run (Run B) continues to change between different spatial solutions even after warm-up. The collected data therefore do not support the conclusion that increasing the warm-up duration alone would resolve the instability.

7.2 Acoustic-to-Visual Projection

The acoustic overlay is generated from the MUSIC azimuth/elevation estimate. Projection error is the pixel distance between that overlay and the manually clicked visual GT. A large error can therefore result from acoustic angular error, projection-calibration error, grid-boundary saturation, or the manually selected visual point not being exactly coincident with the true acoustic-emitting part of the covered source.

The captured screenshots also show cases where the acoustic heatmap visually surrounds the covered source. The absorber-controlled LEFT result (Run A) is a strong qualitative demonstration that the system places acoustic energy on the correct side of the camera view even when YOLO is distracted by other objects in the scene.

7.3 Vision (YOLO) Results

YOLO11n segmentation loaded and ran successfully on CPU. Across the recorded frames it reported chair, person, mouse, TV, laptop, and other objects. Because the actual acoustic source was covered and the room contained many salient objects, semantic detection was inconsistent from frame to frame. This behaviour should be recorded separately from acoustic localisation performance, since the two subsystems are evaluated against different ground truths.

8. Experimental Artifacts

The following files were produced during this validation phase:

- scratch\left_final_validation_representative.jpg
- scratch\left_final_validation_raw.jpg
- scratch\center_final_validation_representative.jpg
- scratch\center_final_validation_raw.jpg
- scratch\final_integrated_validation_results.json

Archival note: the validation script reused the filenames center_final_validation_representative.jpg and center_final_validation_raw.jpg across multiple manually selected source positions. When archiving these images, rename or copy them using the actual GT coordinate or run name so their provenance is unambiguous.

9. Conclusion and Future Work

This experimental phase has demonstrated the intended operation of the real-time multimodal acoustic camera. The complete acquisition and processing chain runs on the actual hardware, and the experiments show a consistent directional response from the UMA-16 array together with the ESP32-CAM view.

Results should be read together with the actual ground-truth clicks and captured images. Some console headings were left on the wrong preset during testing, which makes individual run reports appear to fail when read in isolation; the underlying measurements and images confirm that the system itself is functioning, and the incorrect headings are experimental bookkeeping errors rather than system faults.

The core implementation and functional validation are complete. The main remaining improvement is final pixel-level calibration of the acoustic direction onto the camera image. This can be addressed through more accurate camera calibration, a more rigid camera/array mounting arrangement, and further projection refinement, and does not require rebuilding the core acoustic or multimodal pipeline.